

# Toy Object Parameters Guide

Plain-language reference for the Interactive Object Recovery tool

**Source:** `Foreground Masking/interactive object recovery.py`

**Short answer:** The Toy object model now has three available profiles: Gaussian star, Star cluster, and Compact galaxy. Location, brightness, FWHM, and truth dilation apply broadly. Axis ratio and object PA apply only to Compact galaxy.

## Object Types

| Type           | Profile used                                        | Best interpreted as                                                             | Important notes                                                                                                                                  |
|----------------|-----------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Gaussian star  | Single circular 2D Gaussian profile.                | A compact point-source test object with a smooth, symmetric core.               | Characterised by centroid, brightness, and FWHM. Uses FWHM to set Gaussian sigma. Ignores axis ratio and object PA.                              |
| Star cluster   | Blend of three offset circular Gaussian components. | A small unresolved or barely resolved clump rather than a single isolated star. | Characterised by one centroid, brightness, and FWHM. FWHM sets the scale for all component offsets and widths. Ignores axis ratio and object PA. |
| Compact galaxy | Single elliptical Gaussian profile.                 | A small galaxy-like contaminant with an elongated shape.                        | Characterised by centroid, brightness, major-axis FWHM, axis ratio, and object PA.                                                               |

## Parameter Glossary

| Parameter                            | Meaning                                                                                                       | Applies to                                         | Practical note                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Type                                 | Chooses the mathematical profile used to generate the injected Toy object.                                    | All types                                          | The internal values are gaussian, cluster, and galaxy.                                                                               |
| Brightness                           | Chooses how the object brightness is set: either by peak residual sigma or by integrated magnitude.           | All types                                          | Peak residual sigma is usually the simpler test mode. Integrated magnitude is useful when the FITS photometric calibration is valid. |
| x deproj [arcsec], y deproj [arcsec] | Position of the Toy object in the deprojected, bar-aligned coordinate system.                                 | All types                                          | The code converts this deprojected position back to observed image pixels before injecting the model.                                |
| peak [resid sigma]                   | Peak brightness expressed as a multiple of the robust residual-image noise.                                   | All types, when Brightness is Peak residual sigma  | Example: 30 means the model peak is 30 times the robust residual sigma.                                                              |
| integrated mag                       | Target total magnitude for the whole injected object, not just the central pixel.                             | All types, when Brightness is Integrated magnitude | The code scales a unit-peak model so its integrated flux matches this magnitude.                                                     |
| zero flux [Jy]                       | Flux density of a zero-magnitude source, in Jansky.                                                           | Magnitude mode                                     | The default 280.9 Jy is appropriate for Spitzer/IRAC 3.6 micron Vega magnitudes. Change it for another band or magnitude system.     |
| FWHM [arcsec]                        | Full width at half maximum: the diameter across the profile where the brightness has fallen to half the peak. | All types                                          | Converted to pixels using the galaxy pixel scale. For Gaussian profiles, $\sigma = \text{FWHM} / 2.3548$ .                           |
| axis ratio                           | Minor-axis width divided by major-axis width. A value of 1.0 is circular; smaller values are more elongated.  | Compact galaxy only                                | In the current code, Gaussian star and Star cluster are circular and ignore this value.                                              |

| Parameter           | Meaning                                                                               | Applies to          | Practical note                                                                                     |
|---------------------|---------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------|
| object PA [deg]     | Position angle of the object's major axis, measured in image-pixel coordinates.       | Compact galaxy only | Only matters when the model is elliptical. It is not used for circular profiles.                   |
| truth dilation [px] | Extra pixel dilation applied to the truth mask after thresholding the injected model. | All types           | This changes the evaluation mask used for recall/overlap, not the injected object's light profile. |

## Applicability Matrix

| Parameter                | Gaussian star  | Star cluster   | Compact galaxy |
|--------------------------|----------------|----------------|----------------|
| Type                     | Yes            | Yes            | Yes            |
| Brightness mode          | Yes            | Yes            | Yes            |
| x/y deprojected location | Yes            | Yes            | Yes            |
| peak residual sigma      | Yes            | Yes            | Yes            |
| integrated mag           | Yes            | Yes            | Yes            |
| zero flux [Jy]           | Magnitude mode | Magnitude mode | Magnitude mode |
| FWHM [arcsec]            | Yes            | Yes            | Yes            |
| axis ratio               | No             | No             | Yes            |
| object PA [deg]          | No             | No             | Yes            |
| truth dilation [px]      | Yes            | Yes            | Yes            |

## Key Terms and Formulas

### Gaussian sigma and FWHM

A Gaussian profile is often described by sigma, but the UI asks for FWHM because it is easier to interpret visually. The code uses:

**Formula:**  $\text{sigma\_pixels} = \max(0.2, \text{FWHM\_pixels} / 2.3548)$

A larger FWHM makes the object broader. For the Star cluster profile, FWHM also sets how far apart the three Gaussian components are placed.

### How the three object types are characterised

Each Toy object type is deliberately simple. The aim is to test recovery behaviour with controlled source shapes, not to fit a complete astrophysical model.

- Gaussian star: a single circular Gaussian. It is described by its deprojected position, brightness, and FWHM. It has no elongation or orientation.
- Star cluster: a fixed three-Gaussian blend. It is described by one central position, one brightness scale, and one FWHM scale; the component offsets and relative intensities are fixed by the tool.

- Compact galaxy: a single elliptical Gaussian. It is described by position, brightness, major-axis FWHM, axis ratio, and position angle.

## Integrated magnitude mode

Magnitude mode first builds the chosen object profile with peak = 1, then scales that unit model so the total integrated flux matches the requested magnitude.

**Formula:**  $\text{flux\_Jy} = \text{zero\_flux\_Jy} * 10^{(-0.4 * \text{magnitude})}$

If the FITS header uses BUNIT=MJy/sr, the code converts summed model intensity to Jy using the pixel area in steradians. If not, it looks for a count-based magnitude zero point in MAGZP, MAGZERO, or ZEROPOINT.

## Truth mask and recovery metrics

The truth mask is not the same as the light profile. The code thresholds the model at 8 percent of the peak or 8 percent of the maximum model value, then optionally dilates that binary mask by truth dilation [px].

- Recall is recovered truth pixels divided by total truth pixels.
- Incremental recall uses only the new mask pixels produced after injecting the Toy object.
- Incremental precision asks what fraction of the new mask pixels overlap the truth mask.

## Suggested Usage

- Use Gaussian star for a simple compact-source sensitivity test.
- Use Star cluster when you want a blended compact contaminant rather than a single source.
- Use Compact galaxy when elongation and orientation matter; this is the only Toy object type that uses axis ratio and object PA.
- Use Peak residual sigma for quick recovery experiments. Use Integrated magnitude when you want physically calibrated brightness and the FITS header supports the conversion.
- Remember that truth dilation changes the scoring mask, not the injected object itself.